The Fermi-edge double-scaling limit of the quantum Toda chain: small-amplitude Boussinesq soliton scattering

Project *quantum_boussinesq_solitons*, Phase 2d

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Abstract

We carry out the Fermi-edge double-scaling limit of the quantum Toda Bethe ansatz: $\beta \to 0$ with $y = \beta^2/\hbar_B$ fixed. This gives the *exact* quantum phase of small-amplitude Boussinesq soliton scattering, which we compare with the Phase 2c candidate S-matrix **(d)** and with the semiclassical phase.

- *Head-on:* the exact phase is $4\sqrt{y_1 y_2} = \delta/\hbar$, within 10^{-4}. In this corner the candidate's Gamma ratio is in its Stirling regime, so the candidate agrees. A constant Maslov phase cannot be detected here.
- *Overtaking:* the candidate agrees with the exact phase to $\lesssim 10^{-3}$ for $\eta = y_1 - y_2 \gtrsim 2.5$. Near threshold it is much closer than the semiclassical phase: for example, error -0.076 against -0.33 at $(y_1, y_2) = (1.5, 1)$. A residual of 10^{-2}–10^{-1} remains for $\eta \lesssim 1$. It is converged in $\hbar$ and depends on both y_1 and y_2. It is therefore a genuine two-body correction: the candidate is *not* exact in the corner. **(c)** (numerical)

1 Setting and what is computed

We use the conventions of `toda_route.pdf` (Toda chain with spacing a, $b = e^{-a/2}$) and of `report.pdf` (bad Boussinesq equation, $k = \sinh\beta$).

The corner. Boussinesq solitons with $\beta \to 0$ are the continuum limit of Toda solitons with $\kappa = \sqrt{3}\beta$. The actions are related by $S_{\text{Toda}} = 3b\, S_{\text{Bouss}}$, so $\hbar_T = 3b\,\hbar_B$. The semiclassical phases are $O(\beta^2)$: overtaking $\delta \approx (\beta_1^2 - \beta_2^2)\log A$, head-on $\delta \approx 4\beta_1\beta_2$. The exact S-matrix in the corner is therefore a function of the scaling variables

$$y_j = \beta_j^2/\hbar_B,$$

obtained as $\hbar_T \to 0$ at fixed y_j.

Remark on the candidate's scalings. `smatrix_ansatz.pdf`, item 2 of its final list, states that $\eta = O(1)$ needs $\hbar \propto \beta^4$ for overtaking. Since $\ell_1 - \ell_2 \approx \frac{9}{2}(\beta_1^2 - \beta_2^2)$, one has $\eta \approx (\beta_1^2 - \beta_2^2)/\hbar_B = y_1 - y_2$, so $\hbar \propto \beta^2$ suffices. For head-on scattering $\eta \approx \frac{4}{9}/\hbar_B \to \infty$ in the corner, so the head-on Gamma ratio *is* in its Stirling regime there. Then the candidate reduces to $e^{i\delta/\hbar - i\pi/2}$. **(c)** (elementary)

What is computed. We solve the finite-$\hbar$ asymptotic Bethe ansatz of the quantum Toda chain, with exact digamma kernel, Nyström method, LU reuse and a secant search for the Fermi edge (`scripts/toda_edge_ba.py`).

- Solitons are fixed by their *physical dressed momenta*, $P_{\text{dr}}(p) - P_{\text{dr}}(q) = P_n(\kappa)/a$. This avoids any convention in the bare quasi-momentum.

- For head-on scattering we form the normalised label-frame phase

$$\Psi_T = \big[\Phi(1|2) - \Phi(1|e_2) - \Phi(e_1|2) + \Phi(e_1|e_2)\big] - M_1\big(P_{n2}/a - \Pi_2\big),$$

  where e_j is the edge excitation. This vanishes when either soliton shrinks to the Fermi edge, which removes all single-particle conventions, including constant phases. The last term is the classical frame change (`toda_route.pdf`, Prop. 2).

- For overtaking the two-term version of §2 is used. We compare $\Psi_T/\hbar_T$ with the candidate's phase normalised in the same way, and with the semiclassical $\delta/\hbar_B$.

2 Results

Normalisation. The classical relation $\Phi(1|2) = \delta_T + M_1(P_{n2}/a - \Pi_2) + f(\kappa_1)$ was verified only for colliding configurations, $\kappa_1 > \kappa_2$. We therefore remove only $f(\kappa_1)$:

$$\Psi_T = \Phi(1|2) - \Phi(1|e_2) - M_1(P_{n2}/a - \Pi_2),$$

and normalise the candidate in the same way, $\arg S(1,2) - \arg S(1,e_2)$ (`scripts/toda_edge_twoterm.py`).

A failure along the way. A first, four-term normalisation also subtracted $\Phi(e_1|2)$. That configuration has a slower soliton 1, which never collides, and there the classical relation does not hold. This normalisation produced spurious $O(1)$ differences, for example -2.04 (`scripts/toda_edge_scan.py`, `scripts/toda_edge_grid.py`). It also contained an omitted $S(e,e)$ term for the candidate, which produced an apparent $\pi/2$ in the head-on channel. Both are recorded here and superseded.

Convergence. We used $\hbar_T = 0.01, 0.005, 0.0025$ (N up to 6397 nodes). The values change by $< 4 \times 10^{-3}$ between successive $\hbar$, and we extrapolate quadratically.

(y_1, y_2)	$\eta \approx y_1 - y_2$	exact (Toda)	candidate	semiclassical	exact−candidate
$(1.5, 0.5)$	1.0	-2.7009	-2.6949	-2.6339	-0.0060
$(3, 0.5)$	2.5	-4.3477	-4.3463	-4.3351	-0.0014
$(6, 0.5)$	5.5	-6.5400	-6.5392	-6.5366	-0.0008
$(1.5, 1)$	0.5	-2.6215	-2.5451	-2.2924	-0.0764
$(3, 1)$	2.0	-5.3134	-5.2962	-5.2678	-0.0172
$(6, 1)$	5.0	-8.6817	-8.6757	-8.6701	-0.0059
$(0.3, 0.1)$	0.2	-0.6129	-0.7044	-0.5268	$+0.0914$
$(3, 0.1)$	2.9	-2.1425	-2.1438	-2.1419	$+0.0013$

All columns are quadratic extrapolations $\hbar_T \to 0$ of values computed at the same (y_1, y_2) and $\hbar$. The candidate and semiclassical phases are evaluated at the finite $\beta_j = \sqrt{y_j \hbar_B}$ of each run. See `scripts/toda_edge_twoterm.json` for all raw numbers.

Head-on. With the four-term normalisation, which is valid here because both configurations collide, the exact phase extrapolates to 2.00000, 4.00002 and 4.00000 at $(y_1, y_2) = (0.5, 0.5), (1, 1), (2, 0.5)$. These equal $4\sqrt{y_1 y_2}$. The candidate, correctly normalised, agrees (`scripts/toda_edge_scan.py`).

3 Conclusions

1. In the small-amplitude corner, the candidate captures the exact two-body phase far better than semiclassics. The Gamma function in η carries most of the threshold structure. **(c)**

2. It is not exact. A residual two-body correction of $O(10^{-1})$ remains at $\eta \lesssim 1$. Its sign changes between $(1.5, 1)$ and $(0.3, 0.1)$, so it is not a single CDD factor of definite sign in η alone. Since the residual is not a function of η alone, the variable in the exact Gamma function may be the bare Toda quasi-momentum difference $(p_1 - p_2)/\hbar_T$ of the *quantum* excitations. That differs from η by quantum dispersion corrections of relative size $O(1/y)$. This is the natural next test. **(d)**

3. *The quasi-momentum hypothesis of item 2 is refuted* (`scripts/toda_edge_kappa.py`, extrapolated from $\hbar_T = 0.005, 0.0025$). We replaced η by $\kappa = (p_1 - p_2)/\hbar_T$ of the quantum excitations at the same physical momenta. Then:

   - $\kappa - \eta$ is -0.03 to -0.45, and does not vanish at large y (e.g. -0.45 at $(3, 0.1)$);
   - with κ in the Gamma ratio only, the residuals are 0.17–0.86;
   - with κ in the Gamma ratio and the power factor, they are 0.28–0.49;
   - the original candidate's residuals are ≤ 0.11 (a new point $(1.2, 1)$, $\eta = 0.2$, gives -0.11).

   So η, not the bare Toda quasi-momentum, is the right variable. The residual is a separate two-body correction, concentrated at threshold, whose form is still open. **(c)** (numerical)

4. The head-on corner cannot decide the $\pm\pi/2$ Maslov question raised by the defect test: there the constant is a convention.

5. Caveats: the frame-change term is used in its classical form, and the comparison is only in the corner $\beta \to 0$.